

Further Mechanics 1

Notes · Edexcel A Level Further Mathematics — Further Mechanics 1 · Chapter 3: Elastic Strings and Springs

Chapter 3: Elastic Strings and Springs

Why this chapter matters

Why this matters. Hooke's law introduces a *variable* force for the first time in this unit — tension changes as extension changes — which is exactly why energy methods (Chapter 2) become essential rather than optional. This chapter is also a direct stepping stone to elastic collisions (Chapters 4–5), where elastic potential energy sometimes appears in energy-conservation checks.

Key vocabulary

Term	Meaning
Natural length (l_0)	The length of a string or spring when no force is applied to it
Extension (x)	The amount by which a string or spring is stretched beyond its natural length
Compression	For a spring only, the amount by which it is shortened below its natural length (strings cannot be compressed — they simply go slack)
Modulus of elasticity (λ)	A constant, measured in newtons, describing the stiffness of a particular string or spring
Elastic potential energy (EPE)	The energy stored in a stretched or compressed elastic string/spring
Slack/taut	A string is taut when stretched beyond its natural length (tension acts); it is slack (tension = 0) when its length is less than or equal to its natural length

Key formula table

Quantity	Formula & notes
Hooke's law	$T = \lambda x / l_0$ — T = tension (or thrust for a spring), x = extension (or compression), l_0 = natural length
Elastic potential energy	$EPE = \lambda x^2 / (2l_0)$ — valid for extension or compression of a spring; for a string, only valid while taut ($x \geq 0$)
Work done stretching from x_1 to x_2	$W = \lambda(x_2^2 - x_1^2) / (2l_0)$ — equals the gain in EPE, since T is not constant

Concept: Hooke's law

For a string or spring of natural length l_0 and modulus of elasticity λ , stretched or compressed by amount x , the tension (or thrust) is:

$$T = \lambda x / l_0$$

A **string** can only be taut ($x > 0$, so it can only pull, never push) or slack ($x \leq 0$, tension is zero — a slack string exerts no force). A **spring** can be stretched (tension pulls inward) or compressed (thrust pushes outward), and Hooke's law applies with the same formula in both cases, with x taken as positive for extension and “negative extension” (or a separately labelled compression) for compression.

Connecting to $F = kx$. If you have met Hooke's law before in the form $F = kx$, then $k = \lambda/l_0$ is just the “spring constant” wearing different clothes. The modulus of elasticity λ is a property of the material or wire itself (it does not change if you cut it shorter), whereas k depends on the actual length l_0 of the piece you are using. A-level mechanics works with λ and l_0 separately, rather than a single constant k , precisely so you can compare different lengths cut from the same elastic material.

Worked example 1 — basic application of Hooke's law
EXAMPLE 3.1

Question. A light elastic spring has natural length 0.8 m and modulus of elasticity 40 N. One end is fixed and a particle of mass 2 kg hangs in equilibrium from the other end. Find the extension of the spring. (Take $g = 9.8 \text{ m s}^{-2}$.)

Step 1. The particle is in equilibrium, so the tension in the spring equals its weight:

$$T = mg = 2 \times 9.8 = 19.6 \text{ N} \quad \text{M1}$$

Step 2. Apply Hooke's law, $T = \lambda x / l_0$, with $\lambda = 40$, $l_0 = 0.8$:

$$19.6 = 40x / 0.8 \quad \text{M1}$$

$$19.6 = 50x \implies x = 0.392 \text{ m} \quad \text{A1}$$

Answer: the extension is 0.392 m.

EXAM TIP

Remember that it is the equilibrium condition (resultant force = 0) that supplies the first equation in almost every static Hooke's law problem — if you have only ever used $F = ma$ dynamically, it is easy to forget to write “tension = weight” without being prompted. A one-line free-body diagram habit fixes this permanently.

Concept: Elastic potential energy

Because tension increases linearly with extension ($T = \lambda x / l_0$), the work done in stretching a string/spring from natural length to extension x is the area under the T – x graph (a triangle), giving:

$$EPE = \lambda x^2 / (2l_0)$$

This energy is “stored” and can convert into kinetic or gravitational potential energy when the string/spring is released — exactly analogous to GPE, but now the store is elastic rather than gravitational. This is the same formula as the familiar $\frac{1}{2}kx^2$ from GCSE/Physics, just written with k replaced by λ/l_0 — substitute $k = \lambda/l_0$ into $\frac{1}{2}kx^2$ and you get $\frac{1}{2}(\lambda/l_0)x^2 = \lambda x^2 / (2l_0)$.

Worked example 2 — work done stretching a spring (variable force)
EXAMPLE 3.2

Question. A light spring has natural length 0.5 m and modulus of elasticity 60 N. Find the work done in stretching the spring from an extension of 0.1 m to an extension of 0.3 m.

Step 1. Because tension is not constant during the stretch, use the work-done formula (equivalently, the difference in EPE at the two extensions), not $W = (\text{average } T) \times (\text{change in } x)$ unless justified:

$$W = \lambda(x_2^2 - x_1^2) / (2l_0) \quad \text{M1}$$

Step 2. Substitute $\lambda = 60$, $l_0 = 0.5$, $x_1 = 0.1$, $x_2 = 0.3$:

$$W = 60(0.3^2 - 0.1^2) / (2 \times 0.5) = 60(0.09 - 0.01) / 1 = 60 \times 0.08 = 4.8 \text{ J} \quad \text{M1A1}$$

Answer: the work done is 4.8 J.

EXAM TIP

This is the moment to connect back to Chapter 2: ask yourself why you can't just use "average tension $\times$ extra extension" as a shortcut. In fact that shortcut *does* work here because tension is linear in x (so the average of T at x_1 and x_2 , times $(x_2 - x_1)$, gives the same answer) — but it is worth deriving the standard EPE formula from first principles (area under a T - x graph) at least once, so you understand why the formula has an x^2 in it, rather than memorising it as an isolated fact.

Worked example 3 — energy conservation with a spring/string (vertical motion)
EXAMPLE 3.3

Question. A particle of mass 0.5 kg is attached to one end of a light elastic string of natural length 1.2 m and modulus of elasticity 24 N. The other end of the string is fixed to a point O. The particle is held at O and released from rest, falling vertically. Find the speed of the particle at the instant the string first becomes taut plus a further 0.3 m of extension (i.e. total distance fallen = 1.5 m), given the string first becomes taut after falling 1.2 m. (Take $g = 9.8 \text{ m s}^{-2}$.)

Step 1. From release to the string becoming taut (falls 1.2 m), the particle is in free fall (string slack, exerts no force). Use energy conservation, or $v^2 = u^2 + 2as$ with $u = 0$, $a = g$, $s = 1.2$:

$$v^2 = 2(9.8)(1.2) = 23.52 \text{ (this is the KE-equivalent speed}^2 \text{ when the string becomes taut)} \quad \text{M1A1}$$

Step 2. From the string becoming taut to a further extension of 0.3 m, use energy conservation including EPE. Loss in GPE + loss in KE-store = gain in EPE, i.e.:

$$\frac{1}{2}mv_1^2 + mg(0.3) = \frac{1}{2}mv_2^2 + \lambda x^2 / (2l_0) \quad \text{M1}$$

where $v_1^2 = 23.52$ (speed² when taut), the particle falls a further 0.3 m gaining GPE-loss = $mg(0.3)$, and gains EPE = $\lambda(0.3)^2 / (2 \times 1.2)$ with $\lambda = 24$:

$$\text{EPE} = 24(0.09) / 2.4 = 0.9 \text{ J} \quad \text{A1}$$

Step 3. Substitute $m = 0.5$:

$$\frac{1}{2}(0.5)(23.52) + 0.5(9.8)(0.3) = \frac{1}{2}(0.5)v_2^2 + 0.9 \quad \text{M1}$$

$$5.88 + 1.47 = 0.25v_2^2 + 0.9$$

$$7.35 - 0.9 = 0.25v_2^2 \implies v_2^2 = 25.8 \implies v_2 = 5.08 \text{ m s}^{-1} \text{ (3 s.f.)} \quad \text{A1}$$

Answer: the particle's speed at that point is approximately 5.08 m s^{-1} .

EXAM TIP

This three-phase structure (free fall while slack $\rightarrow$ taut with EPE building up $\rightarrow$ possible oscillation) is the classic Further Mechanics 1 “string on a particle” scenario, and it is worth drawing as a vertical number line marked with the natural length point and showing exactly where the string is slack versus taut before attempting any calculation. The single most common error is applying the EPE formula over the *whole* fall (including the slack phase), which silently invents energy that was never actually stored.

Chapter summary

Term	Meaning
Hooke's law	$T = \lambda x/l_0$; strings only pull ($x \geq 0$), springs can push or pull
Elastic potential energy	$\text{EPE} = \lambda x^2/(2l_0)$, stored while stretched/compressed
Work done stretching	equals the change in EPE between two extensions, since T is variable
Energy conservation with elastic elements	$\text{KE} + \text{GPE} + \text{EPE}$ is conserved when no other forces do work; a slack string stores no EPE and exerts no tension

Common exam mistakes

- Applying Hooke's law or the EPE formula to a slack string (must have $x \geq 0$ for a string).
- Using natural length instead of extension (or vice versa) inside $T = \lambda x/l_0$.
- Treating tension as constant when finding work done, rather than using the EPE-based formula for a variable force.
- Forgetting to include GPE changes alongside EPE changes in vertical elastic-string/spring problems.
- Sign errors distinguishing extension from compression in spring problems.
- Mixing units for l_0 and x (e.g. l_0 given in metres but x given in centimetres) — always convert to the same unit, usually metres, before substituting into $T = \lambda x/l_0$.

You should now be able to

- State and apply Hooke's law, $T = \lambda x/l_0$, distinguishing strings from springs.
- Calculate elastic potential energy and use it in energy-conservation equations.
- Find the work done in stretching or compressing an elastic string/spring between two extensions.
- Analyse multi-phase motion involving a particle falling freely before a string becomes taut, then decelerating under tension.